DAY 32 | PRE-READ

Quality Checks Before Every Release

A light read before Day 32. About 15 minutes, no code required.

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Why read this?

Yesterday you measured how good an AI system is. Today you make those measurements automatic, so that every change to a prompt, model or setting is checked for quality and cost before it reaches users.

1. The answer key

Teachers mark exams against an answer key that experts have checked. A **golden dataset** is the answer key for an AI system: questions with verified answers, kept safe and updated when policies change.

2. Small changes, surprising effects

Changing one word in a prompt, or switching to a cheaper model to save money, can quietly break answers that worked yesterday. **Regression testing** reruns the golden questions after every change and flags anything that got worse, just like the tests you wrote for code.

Key idea: treat prompts like code. Version them, test them, and review changes.

3. Did we fetch the right pages?

Measure	Question	Everyday version
Precision	How much of what we fetched was useful?	Of 10 books brought, how many were relevant?
Recall	How much of what we needed did we fetch?	Of the 3 books needed, how many arrived?

4. Pressing the right button NEW (v4)

When an AI agent takes an action, the action is the answer. If a customer asks for a refund of 300 on order 77, the agent must call the refund tool with exactly that order and amount. **Tool-call fidelity** tests check this the way a bank checks every transfer.

5. Good and affordable NEW (v4)

A change that makes answers slightly better but triples the bill is not automatically a win. Quality gates now also check **tokens and cost per question**, so nobody is surprised by the invoice.

6. Think about these before class

- Who in a telecom company could verify golden answers for a support assistant?
- What change might someone make to save cost that could quietly reduce quality?
- Can you think of two correct answers to a question that share almost no words?

Kihon (基本): the foundation, the basics